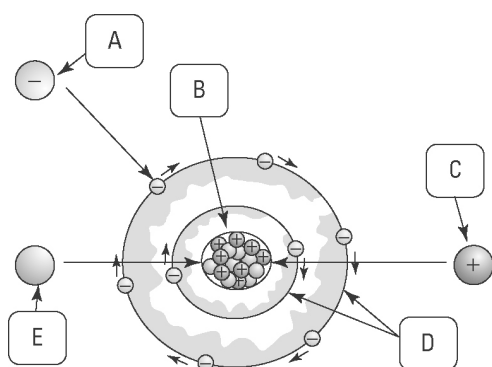


The structure of the atom

Answers

The atom

1. The following drawing shows a simplified structure of an atom.

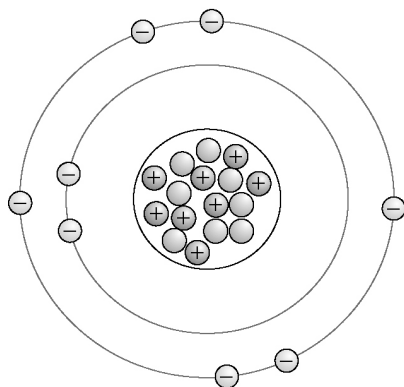


Identify the labelled components of the atom.

- A: Electron
- B: Nucleus
- C: Proton
- D: Electron shells
- E: Neutron

The oxygen atom

2. The diagram below shows the atomic structure of oxygen.



- (a) How many of each type of sub-atomic particle are present?

Electrons: 8

Protons: 8

Neutrons: 8

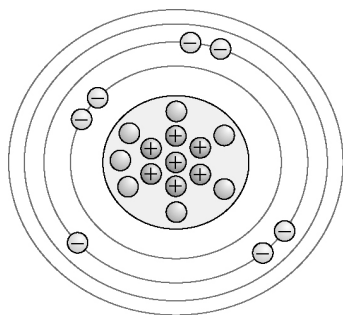
- (b) Is the oxygen atom shown above neutral or charged? Explain.

Neutral — there is an equal number of protons and electrons.

- (c) How many nucleons (total particles) are present in the nucleus? 16

Atoms and ions

3. The diagram below shows a nitrogen atom. A nitrogen atom can form an ion by gaining three more electrons, so that its electron configuration becomes 2, 8.



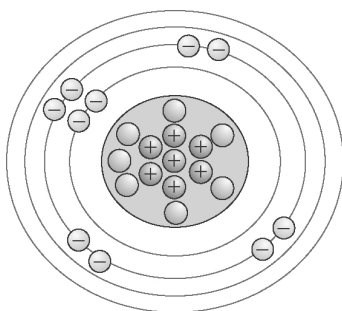
- (a) In the diagram above, how many of each type of sub-atomic particle are present?

Electrons: 7

Protons: 7

Neutrons: 7

- (b) Draw the nitrogen ion described above.



- (c) In the diagram you have just drawn, how many of each type of sub-atomic particle are present?

Electrons: 10

Protons: 7

Neutrons: 7